Group A

Answer All 5 Questions (5 x 2 = 10)

		Knowledge Level	Course Outcome
1	What is odd function?	L1	CO1
2	What is entire function?	L1	CO2
3	Find the value of the Wronskian of the functions $y_1 = e^{4x}$ and $y_2 = xe^{4x}$. Hence conclude about the independency of the functions. [1.5+0.5]	L2	CO3
4	State Gauss divergence theorem.	L1	CO4
5	Find the second order differential equation corresponding to the following simultaneous linear first order differential equation with constant coefficient: $$\frac{dx}{dt} = 3x + 8y$$ $$\frac{dy}{dt} = -x - 3y$$	L1	CO3

Group B

Answer All 5 Questions (5 x 4 = 20)

6	Find $\int_C \frac{4-3z}{z(z-1)(z-3)}\,dz$, $C: \lvert z\rvert = \frac{5}{2}$.	L1	CO2
7	Find the analytic function $f(z) = u + iv$, where $v = \log(x^2 + y^2) + x - 2y$.	L1	CO2
8	Solve the following initial valued problem: $$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 13y = 0, y(0) = 3, \frac{dy}{dx}(0) = -1$$	L3	CO3
9	If $A = (2y+3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$, then evaluate $\int A \cdot dr$ along the curve C given by $x = 2t^2, y = t, z = t^3$ from $t = 0$ to $t = 1$.	L3	CO4
10(a)	Find $\int_C \frac{e^{2z}}{(z-1)(z-2)}\,dz$, $C: \lvert z\rvert = 3$.	L2	CO2
	OR		
10(b)	Solve the following differential equation: $(D^2 - 1)y = 2x^2 - 3x + 1$, where $D = \frac{d}{dx}$	L3	CO3

Group C

Answer All 2 Questions (2 x 10 = 20)

11(a)	i) Solve the following Cauchy-Euler equation: $$4x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 3y = 0$$	L3	CO3

	ii) Find the flux of the vector field $\vec{A} = (x - 2z)\hat{i} + (x + 3y + z)\hat{j} + (5x + y)\hat{k}$ through the upper side of the triangle ABC with vertices at the points $A(1,0,0), B(0,1,0), C(0,0,1)$. (Draw the figure) [4+6]	L1	CO4
	OR		
11(b)	i) Find the general solution of the following equation using the method of variation of parameter: $$(D^2 - 4D + 4)y = e^{-2x}, D = \frac{d}{dx}$$	L1	CO3
	ii) Evaluate the following using Green's theorem $$\oint \{(\cos x \sin y - xy)dx + \sin x \cos y\, dy\}$$ over the circle $C: x^2 + y^2 = 1$. [5+5]	L4	CO4
12(a)	i) Expand the function $f(x) = e^{ax}$ as a Fourier series in the interval $-\pi \le x \le \pi$ ii) If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function of $z = x + iy$, find $f(z)$ in terms of z. [6+4]	L1	CO1 & CO2
	OR		
12(b)	i) Expand the function $f(x) = x - x^2$ as a Fourier series in the interval $-1 < x < 1$ ii) Evaluate $\oint_C \frac{e^z\, dz}{z^2 + \pi^2}$, where C is the circle $\lvert z \rvert = 4$. [6+4]	L1	CO1 & CO2